

Real-Time Pricing: Will a Glowing Orb Lead the Way?

The nation's first large-scale real-time pricing program began in 2003 in the metropolitan Chicago area. It's credited with a 3% decrease in customer energy use in the summer.

by **Marjorie Isaacson and Ruth Klotz-Chamberlin**

A grand experiment is under way for residential electricity customers in Illinois. Consumers are responding with sticker shock at electricity prices after the lifting of a ten-year rate freeze, from 1997 to 2007, and the state is attempting to recover from deregulation fever. Nothing unusual here. But Illinois is unique in enacting legislation that requires large investor-owned utilities to offer real-time pricing rate options to their residential customers.

Real-time pricing (RTP) is a dynamic rate where electricity prices vary hour by hour and day by day to reflect prices set by wholesale electricity markets. Most of the time prices are lower than the flat rate that is usually offered to electricity consumers, but sometimes—usually at times of peak energy use—rates are higher. Overall RTP prices will average out lower than the flat price, but specific days, weeks, and even months can be more expensive. RTP customers who avoid excessive peak energy use are rewarded with even lower electricity costs. And reduction of peak energy use has benefits systemwide that help make

Time-of-Use (TOU), Critical-Peak Pricing (CPP), and Real-Time Pricing (RTP) Rates

The Energy Policy Act of 2005, Section 1252, requires all states (and nonregulated utilities) to consider charging time-based rates, including time-of-use, critical-peak pricing, and RTP. Each of these time-based rates creates incentives for customers to better manage their load, to reduce their use of the most expensive electricity—peak power. Peak-load reduction reduces power costs, reduces stress on the infrastructure, and improves reliability, all of which provide a range of benefits for both utilities and consumers. Most residential programs have utilized CPP or TOU rates, while RTP rates have only been available to commercial and industrial electricity users. Both CPP and TOU have lower off-peak and higher on-peak prices, varying by season and time of day. With CPP, utilities can also ask customers to curtail usage during especially high-load periods. RTP uses hourly prices, which reflect the marginal supply cost to the utility to procure electricity in the wholesale market. Unlike the artificial constructs of block off-peak/on-peak rates, RTP provides accurate price signals that link wholesale and retail markets. In endorsing RTP, economics Nobel laureate Vernon Smith also predicts that “better and cheaper technologies” will be developed to give consumers even more options on variable rates.

the electrical system work better for everyone (see “Time-of-Use, Critical-Peak Pricing, and Real-Time Pricing Rates”).

The two major utilities in Illinois, ComEd and Ameren Illinois utilities, project enrollment of 130,000 customers on RTP rates over the next four years. At that point, the Illinois Commerce Commission, the state's regulatory body, will review the effect of RTP on reducing peak

demand and producing net economic benefits to residential customers and recommend its renewal, modification, or dissolution. Consumers will be evaluating the real-time pricing rate too; in less formal contexts—by looking at their electricity bills, and deciding if the payoff from their attempts to avoid using peak energy was worth the effort. The outcome should provide conclusive evidence as to how much real-time



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utility programs

The Ambient Orb

The Ambient Orb is an example of the transmission of ambient information, where data are displayed in subtle, real-life contexts rather than on a computer screen. Unlike tools such as the The Energy Detective, the Ambient Orb requires little from the consumer. CNT Energy had its orbs programmed to signal the current estimated hourly energy prices, and named them PriceLights. Using a spectrum of colors that ranges from blue to red, the signal changes hourly, to reflect price fluctuations. Time periods where the electricity price exceeded the high-price alert level of \$0.13/kWh were marked with a slow red pulse.

rates can contribute to a more-efficient electrical system.

The nation's first large-scale real-time pricing program began in 2003 in the metropolitan Chicago area. It was developed by CNT Energy (formerly known as the Community Energy Cooperative) in cooperation with ComEd and received funding from the Illinois Department of Commerce and Economic Opportunity. The 1,500 participants in the Energy-Smart Pricing Plan (ESPP) were successful in cutting peak demand in response to price signals. In addition to shifting load, the plan conserved energy, but only during the summer months. The exact evaluation finding was that there was a reduction in use of 27 kWh per month (3%) in June through September, but no reduction in use in other months. The evaluators arrived at this finding by statistical modeling that compared individual customers' actual to expected use.

How the Program Works

In contrast to other pricing pilots, the ESPP program was low-tech in its approach. The only equipment required was the replacement of the standard meter with an interval meter capable of recording hourly energy use. These meters are considerably less expensive than more-sophisticated meters that provide customers with real-time feedback on their electricity use. Using these more sophisticated meters would have greatly increased the cost of the program, and the interval meters were

compatible with ComEd's standard meter setup.

Another main focus of the program was education. Before CNT Energy launched the pilot, it conducted an analysis of electricity prices.

The analysis showed that there were predictable price trends, and that consumers could manage these trends, given adequate education. The information that customers needed to utilize RTP rates successfully was available from a variety of sources. CNT Energy provided ongoing access to real-time prices by phone or via a dedicated Web site. Program materials (a reference guide and periodic newsletters) educated consumers about daily and seasonal patterns of electricity prices. And when prices exceeded a previously established threshold (\$0.13/kWh as of this writing), participants were alerted to watch their electricity use during the relevant hours. This high-price alert was delivered by phone or e-mail the evening before the high prices were scheduled to occur.

Program participants also had password-protected access to a Web site where they could see their electricity use by the month, week, day, or hour, and compare their RTP bills to what they would have paid had they remained on the flat rate.



4¢

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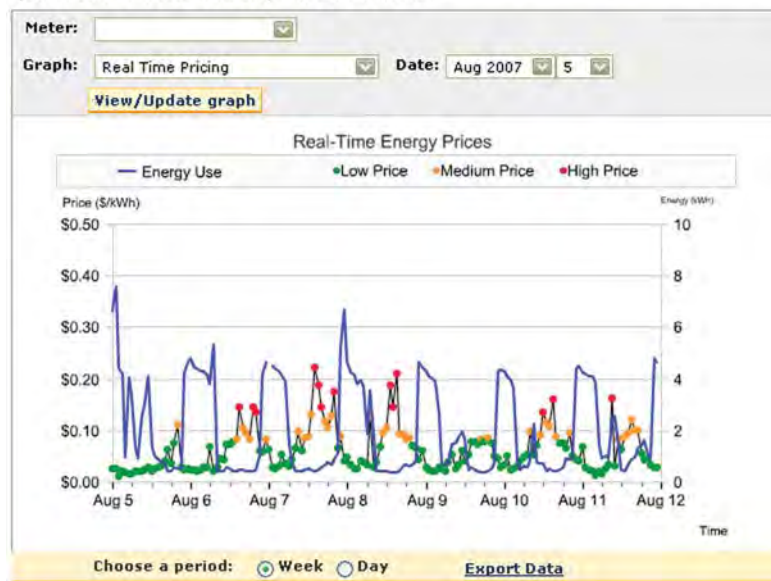
When the Ambient Orb (or Pricelight) pulses red, homeowners are alerted that electricity is expensive.

However, customers were not required to have access to the Internet in order to participate in the program. This enabled customers on the have-not side of the digital divide to participate successfully. Other time-based programs have relied on participants to have Web-connected homes that could link with the smart meters. In contrast, 30% of ESPP participants had no computer in their home, and received their high-price alerts via telephone. These customers included both low-income and minority populations, including a Spanish-speaking subset.

Over the first four years of the ESPP program, participants responded positively, with 80% stating that they found participating in ESPP "quick and easy." Program participants achieved an overall savings in electric bills of 10% over the flat rate. Individual savings varied, with some customers saving 20% and a few even paying more than they would have on the flat rate. Customers were surveyed about the techniques they used to avoid using electricity during the higher-priced hours. They reported behavior changes that

My Energy Usage and Costs

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While customers may have been interested in detailed information about their electricity use, they may not have been interested enough to invest their discretionary time in learning to use and using the Web tools.

ranged from prosaic, such as running the dishwasher or clothes washer at nonpeak times; to creative, such as reinstalling their clothes lines or cooking outdoors; to exotic, such as embracing a nocturnal lifestyle. Customers also reported that participating in the ESPP program inspired them to conserve energy at all times, not just during the high-price alerts or higher-priced periods.

Analysis of the number of customers who utilized the Web site tools showed that relatively few participants took advantage of them. Only 16% ever activated their accounts, and even fewer used the Web site more than once. Most customers seemed satisfied with the twice-yearly paper reports on their savings, delivered by traditional mail along with one of the quarterly newsletters. This finding was unexpected. One interpretation was that although the low-tech approach provided no real-time feedback information the customers preferred it. While customers may have been interested in detailed information about their electricity use, they may not have been interested enough to invest their discretionary time in learning to use and using the Web tools. An alternative explanation was that the

Web tools were not user-friendly or compelling enough to capture the customers' interest. This question is still being analyzed, and it's possible that both explanations are to some extent correct.

In a theoretical context, this finding was problematic. The ESPP program was based on the premise that when customers were provided with good information, they would make smart choices. Was there a way to convey information that was fast, easy, and perhaps even enjoyable?

Enter a Glowing Orb

The Ambient Orb is a small glass globe equipped with a spectrum of colored LED lights (see "The Ambient Orb"). The orb uses 2 watts; at \$0.10/kWh, operating costs are about \$1.75 per year. Powered by a standard electrical outlet, and controlled via a pager signal, orbs can be programmed to change colors that represent data. Originally, the orb was designed to help investors monitor the stock market. Other standard applications include weather predictions and traffic congestion. CNT Energy wasn't the first to use the Ambient Orb to convey electricity pricing information.

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Customers at Southern California Edison who used the orb reduced their peak-period energy use by 40%. Utility Program Development Manager Mark Martinez explains that the orb worked better than other alerts. The glowing sphere was both less annoying and more persistent than other notification methods they had tried, including automated phone calls, text messages, and e-mails.

CNT Energy solicited a subset of participants to receive an orb (branded a PriceLight) in addition to their traditional price notification methods. The participants with PriceLights responded to the orb by using even less electricity during peak times than they had done before they got the orb. Evaluation of customers who were part of the PriceLight study group found an even stronger response to hourly prices, with an elasticity of -0.067 . But the affective response to the PriceLight was even more compelling. Customers were lavish in describing their fondness for the PriceLight. In particular, participants with children praised the PriceLight for providing an accessible, real-world context for teaching their children about energy use. Additionally, the orb's appeal has a lot to do with its attractive design. CNT Energy promoted the orb on TV, and in one of these promotional pieces it was sitting in a display with other round-shaped objets d'art.

One major drawback of the PriceLight is its cost. At \$150 retail for the hardware and a monthly fee for the data feed subscription, PriceLights are prohibitively expensive. The PriceLights were originally distributed at no cost, with only a \$25 refundable deposit. But even while they proclaimed their affection for their orbs, participants said they were willing to pay only an average of \$25 for the product.

Other less flashy devices have proven to be efficient for reducing energy use. CNT offered some participants an option where their central air conditioners would be automatically cycled when prices exceeded the alert level. Programs

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that combine automated controls with price signals have resulted in peak demand reductions that are up to twice as great as reductions from dynamic pricing or load control programs alone. CNT customers also used this option successfully; automatic cycling for air conditioners resulted in higher elasticities for these customers than for other participants in the ESPP program. The cycling option was also popular with participants, with 24% reporting no change in comfort levels, and 62% reporting a decreased, but still satisfactory, level of comfort. (There were five possible responses to this question: Increased comfort level; Decreased comfort level, but I am still satisfied; Decreased comfort level, but I am not satisfied; No change; and Don't know.)

Meanwhile, the PriceLight's photogenic qualities guaranteed that it would play a prominent role in any media coverage of the ESPP program. New applicants often saw the PriceLight as an essential accessory, and insisted that their participation was contingent on their receiving a PriceLight. Exactly how valuable the PriceLight is in terms of reducing energy use is difficult to quantify. If the electricity reductions were large enough, it might pay utilities to subsidize PriceLights for their customers. To reduce costs, Ambient Technologies has developed a less expensive display device called the Energy Joule; this device has not yet been tested as an RTP tool.

Enrollment in RTP programs is ongoing in the territories served by both Ameren Illinois utilities (Power Smart Pricing) and ComEd (Residential Real-Time Pricing). The Illinois Clean Energy Community Foundation has paid for 150 PriceLights, which are being offered to Ameren customers at reduced costs.

The Glow of Success

As electricity prices continue to rise, customers will look for opportunities to control costs. The realities of climate

change also make it imperative to use electricity efficiently. Therefore, variable rates should become a more common option for customers. In some areas, in fact, they are being considered as the default rates for residential consumers.

Most utilities looking at time-varying rates are looking at time-of-use (TOU) or critical-peak pricing (CPP) models, rather than at real-time pricing. Only a few RTP pilots are under way or under consideration. Given the success of RTP in Illinois, it is surprising that this option is not gaining more attention than it is. Or perhaps not. The rate design working group of the National Action Plan for Energy Efficiency has stated that "economists and public policy analysts can become enamored with efficient pricing schemes, but customers generally prefer simple rate forms ... Rate designs that are too complicated for customers to understand will not be effective at promoting efficient consumption decisions." But this assertion seems to sell consumers short. Just as consumers have become skilled at evaluating the intricacies of cell phone rates, the Illinois experience shows that they can navigate the potential volatility of real-time electricity prices.

Real-time pricing offers the best opportunity for consumers to reap the rewards of energy-efficient behavior. The reason we think RTP is best is that often the prices can go even lower than they could on a time-of-use rate. The disadvantage is that there is more fluctuation in the prices and they can peak a bit higher, but over time we are seeing that the value of buying directly from the wholesale market translates into savings. Here's hoping a few more regulators and utilities give consumers the opportunity to use RTP.

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Participants with children praised the PriceLight for providing an accessible, real-world context for teaching their children about energy use.

of the PriceLight project at CNT and now works for the Alliance for Healthy Homes in Washington, D.C.

For more information:

Barbose, Galen, Charles Goldman, and Bernard Neenan. *A Survey of Utility Experience with Real-Time Pricing*, LBNL-54238. Berkeley, California: Lawrence Berkeley National Laboratory, December 2004 (<http://repositories.edlib.org/lbnl/LBNL-54238>).

York, Daniel, and Martin Kushler. *Exploring the Relationship Between Demand Response and Energy Efficiency: A Review of Experience and Discussion of Key Issues*. Washington, D.C.: American Council for an Energy Efficient Economy, 2005.

For a copy of the DOE's national Action Plan for Energy Efficiency, go to www.epa.gov/cleanenergy/energy-programs/napee/index.html.